Dear Editors,

We submit our manuscript, “Warming and survey effort contribute comparably to new bird distribution records when dated to discovery”, for consideration as a Research Article in *Global Change Biology*. Our answers to the journal’s cover-letter questions follow, each in 50 words or fewer.

1. What scientific question is addressed in this manuscript?

When a bird species is recorded in a new province, how much does the record reflect climate-driven range change, and how much the growth of survey effort? And does the usual practice of dating records to their publication year bias the answer?

2. What is/are the key finding(s) that answer this question?

Publication-year dating inflates the survey-effort effect by 37% and creates a spurious climate effect. Dated to discovery, species-referenced warming (hazard ratio 1.36 per SD) and effort (1.43) contribute comparably. Province-by-year survey pulses are the largest variance component, and returns on effort are falling; warming shows no such decline.

3. Why is this work important and timely?

New distribution records are widely read as fingerprints of climate-driven range shifts, yet citizen-science effort has grown as fast as the climate has warmed. Without separating the two, range-shift attribution, monitoring priorities and protected-area assessments risk mistaking the history of observation for ecological change.

4. Describe how your paper fits within the scope of GCB.

GCB promotes understanding of biological responses to environmental change. We quantify one such response, the redistribution of bird species across China under warming, while explicitly modelling how it is detected, and we delimit how far the fitted relationships can be projected under CMIP6 scenarios.

5. What biological AND global change aspects does it address?

Biological: first provincial records of 392 bird species, with species detectability, migratory strategy and traits. Global change: accumulated warming relative to each species’ historical range, CMIP6 SSP2-4.5 and SSP5-8.5 projections, and the rapid growth of citizen-science monitoring that shapes which changes are observed.

6. What are the three most recently published papers that are relevant to this manuscript?

Chen, S. et al. (2025) *Avian Research* 16, 100310 (https://doi.org/10.1016/j.avrs.2025.100310); Bowler, D. E. et al. (2025) *Biological Reviews* 100, 50–67 (https://doi.org/10.1111/brv.13127); Xing, X. et al. (2024) *Frontiers in Ecology and Evolution* 12, 1415268 (https://doi.org/10.3389/fevo.2024.1415268).

7. Provide the name and institutional email addresses of five potential reviewers.

[To be provided by the authors: five researchers without conflicts of interest, with institutional email addresses; for example, specialists in biodiversity-data bias, range-shift detection, and discrete-time hazard or occupancy modelling.]

Related manuscripts. A data descriptor for the China Bird New Record dataset (CBNR) is under consideration at *Scientific Data*; it documents the dataset and contains none of the analyses reported here. Our companion study of new mammal distribution records in China has been published (Ding et al., 2025, *Global Ecology and Biogeography*).

This manuscript has not been published and is not under consideration elsewhere. All authors have approved its submission and declare no conflicts of interest. [Authors: confirm, and state any preprint.]

Sincerely,

Chenchen Ding, on behalf of all authors Institute of Ecology, College of Urban and Environmental Sciences, Peking University, Beijing 100871, China chenchen.ding@pku.edu.cn